

News Comment Topic Analysis System

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A Semester Project

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الملخص

تمثل تعليقات الأخبار مصدرًا غنيًا لفهم تفاعل المستخدمين مع الأحداث السياسية والاجتماعية والاقتصادية والصحية، لكنها تُعد من أصعب أنواع النصوص للتحليل الآلي بسبب طبيعتها القصيرة وغير الرسمية وكثرة الضجيج والسخرية وتداخل المواضيع. ومع ازدياد حجم هذه التعليقات، يصبح التحليل اليدوي غير عملي، كما تصبح الطرق التقليدية غير كافية لاستخراج البنية الدلالية الحقيقية للنقاشات.

يعالج هذا المشروع مشكلة اكتشاف المواضيع الكامنة من تعليقات الأخبار دون الاعتماد على تصنيفات بشرية جاهزة، مما يجعله نظام نمذجة مواضيع غير إشرافي (Unsupervised). وبناءً على ذلك، لا يتم تدريب النظام على فئات محددة مسبقًا، بل يعتمد التقييم على مقاييس جودة هيكلية مثل نسبة الضجيج، عدد التعليقات المسندة، وقوة الثقة في الإسناد وتنوع الكلمات المكتشفة.

يقوم الحل المقترح على تحسين فضاء التمثيلات الدلالية قبل تطبيق BERTopic. تم استخدام نموذج SentenceTransformer مبني على MiniLM، وتدريبه بأسلوب contrastive fine-tuning باستخدام 1,500,000 زوج تدريبي موجب من نوع (Comment ↔ Comment) تم استخراجها تلقائيًا مع اعتماد تقسيم post-level split لضمان نزاهة التقييم ومنع تسرب البيانات.

شملت قاعدة البيانات النهائية 843,636 تعليقًا نظيفًا. استُخدمت التمثيلات الناتجة من النموذج V7 داخل خط معالجة يعتمد على (BERTopic (UMAP, HDBSCAN, c-TF-IDF). أُجري التقييم النهائي على مجموعة اختبار ثابتة تضم 58,014 تعليقًا، حيث حقق النموذج إسناد 39,458 تعليقًا لمواضيع صالحة بنسبة 68.01%، ونسبة ضجيج 31.99%، مع اكتشاف 94 موضوعًا بمتوسط ثقة 82.52%.

أظهرت المقارنة مع نماذج أخرى أن تحسين مؤشرات التدريب لا يضمن نتائج أفضل في التجميع النهائي، وتم اعتماد V7 لتفوقه في تحقيق أفضل توازن بين التغطية وتقليل الضجيج. تؤكد النتائج أن تحسين التمثيلات الدلالية ساعد بشكل كبير في تحسين تماسك البنية الموضوعية المستخرجة من تعليقات الأخبار، مما يوفر أداة تحليلية قوية للبيانات الضخمة.

الكلمات المفتاحية: نمذجة المواضيع، SentenceTransformer، BERTopic، تعليقات الأخبار، تمثيلات دلالية، Contrastive Fine-Tuning، UMAP، HDBSCAN.

ABSTRACT

News comment sections contain vast amounts of short, informal, and noisy user-generated text reflecting public reactions to global events. Analyzing these manually is impractical due to scale and linguistic complexities such as sarcasm and ambiguity. This project addresses the challenge of discovering latent topics without manual labels, employing an **unsupervised topic modeling** approach rather than traditional supervised classification.

The proposed solution enhances the semantic embedding space prior to applying the BERTopic pipeline. A SentenceTransformer model (MiniLM-L12-v2) was fine-tuned using 1,500,000 automatically mined **Comment** $\leftrightarrow$ **Comment** positive pairs. To ensure robust evaluation and prevent data leakage, a post-level split strategy was implemented, maintaining the integrity of the training and testing datasets.

From an initial dataset of over one million records, 843,636 clean comments were utilized. The final adopted model, **V7**, was trained using MultipleNegativesRankingLoss. Its generated embeddings were processed through UMAP for dimensionality reduction, HDBSCAN for density-based clustering, and c-TF-IDF for thematic word representation.

Evaluation on a fixed pool of 58,014 comments showed that V7 achieved 39,458 valid topic assignments (68.01% coverage) and effectively reduced noise to 31.99%. The system detected 94 topics with an average confidence of 82.52% and a topic diversity score of 0.9011, outperforming baseline models in clustering quality and thematic coherence.

Comparisons with experimental variants (V5, V6, V8, V9) demonstrated that embedding-level metrics alone do not guarantee superior topic modeling. V7 was selected for its optimal balance between valid coverage and noise reduction. This work establishes that domain-adapted embeddings are essential for transforming noisy, short-text news discourse into structured, interpretable insights.

Keywords: Topic Modeling, BERTopic, SentenceTransformer, News Comments, Semantic Embeddings, Contrastive Fine-Tuning, UMAP, HDBSCAN, c-TF-IDF.

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CHAPTER 1

RESEARCH INTRODUCTION

This chapter introduces the research context, scientific problem, objectives, contributions, beneficiaries, and organization of the report.

1.1 INTRODUCTION

Online news platforms and social media pages generate large volumes of user comments every day. These comments represent public reactions to political events, economic issues, health-related news, social debates, cultural topics, and international developments. Unlike formal news articles, user comments are usually short, informal, noisy, and context-dependent. They may include sarcasm, repeated expressions, incomplete sentences, personal names, emotional reactions, user mentions, and unrelated side discussions.

Manual analysis of such comments becomes impractical when the dataset contains hundreds of thousands or millions of records. A human analyst can inspect a limited sample, but it is difficult to identify recurring discussion themes reliably across a large-scale comment collection. This creates a need for automatic methods that can organize large groups of comments into meaningful and interpretable topics.

This project addresses the problem of extracting topics from news comments using an unsupervised topic modeling approach. The system does not depend on manually labeled classes such as politics, health, religion, or economy. Instead, it attempts to discover latent topics directly from the comment data. This makes the project fundamentally different from supervised text classification, where each document is trained against a predefined target label.

The proposed system uses a fine-tuned **SentenceTransformer** embedding model followed by a **BERTopic**-based topic modeling pipeline. The embedding model converts comments into semantic vectors, while BERTopic uses these vectors to discover topic clusters through dimensionality reduction, density-based clustering, and topic representation.

The final adopted embedding model is named the **Final Fine-Tuned MiniLM Comment Encoder**, internally referred to as **V7**. The main assumption of the project is that improving the semantic representation of news comments before topic modeling can help reduce noise, increase valid topic assignment, and produce more coherent topic structures.

1.2 SCIENTIFIC PROBLEM

The main scientific problem is the difficulty of discovering meaningful and coherent topics from a large collection of news comments without manually annotated topic labels. In supervised classification, each document has a known target label, and the model learns to predict that label. In this project, however, the dataset does not contain gold-standard topic labels that define the correct topic of each comment.

Because no manually labeled topic classes are available, the project cannot be treated as a supervised classification task. Final evaluation metrics such as accuracy, precision, recall, and F1-score are not suitable as primary project metrics. Instead, the system must be evaluated using unsupervised topic modeling indicators such as valid topic comments, noise comments, noise ratio, detected topics, confidence scores, topic diversity, and manual topic inspection.

News comments introduce additional technical challenges. Many comments are short and may contain only a few words. Some comments are sarcastic, ambiguous, emotional, or unrelated to the main news event. Others contain generic reactions such as agreement, surprise, anger, jokes, or personal remarks. These properties make it difficult for a topic modeling system to distinguish coherent discussion themes from noise.

Another methodological challenge is data leakage. Comments under the same news post often share a common contextual meaning. If comments are split randomly at the comment level, comments from the same discussion may appear in both training and testing sets. This may cause the model to indirectly learn discussion-specific patterns and produce overly optimistic evaluation results. For this reason, the project uses a post-level split strategy.

The scientific problem can be summarized as follows: how can an unsupervised topic analysis system discover coherent latent topics from large-scale news comments without manually labeled topic classes, while reducing noise and improving the quality of topic assignments?

A general pretrained embedding model may not represent the specific semantic structure of noisy news comments sufficiently well. Therefore, this project investigates whether fine-tuning a SentenceTransformer model on automatically mined **Comment** ↔ **Comment** positive pairs can improve downstream BERTopic performance.

1.3 RESEARCH OBJECTIVES

The main objective of this project is to build an unsupervised topic modeling system for analyzing large-scale news comments using fine-tuned semantic embeddings and a BERTopic-based pipeline.

The specific objectives of the project are:

1. To prepare and clean a large collection of news comments for semantic embedding generation and topic modeling.
2. To preserve a post-level split in order to reduce data leakage between training, validation, and testing.
3. To construct automatically mined positive pairs of type **Comment** ↔ **Comment** from news discussion threads.
4. To fine-tune a SentenceTransformer model using the mined positive pairs rather than manually annotated topic labels.
5. To generate semantic embeddings for news comments using the final fine-tuned embedding model.
6. To apply BERTopic using UMAP for dimensionality reduction, HDBSCAN for clustering, and c-TF-IDF for topic representation.
7. To evaluate the discovered topics using appropriate unsupervised topic modeling metrics instead of supervised classification metrics.
8. To compare the final model with previous and additional project models, including V5 Stage 1 Controlled, V6-B, V8, and V9 BGE where verified results are available.
9. To perform manual topic inspection in order to distinguish strong, mixed, and weak topics.
10. To document the complete development process, including data preparation, pair mining, model training, evaluation metrics, final results, limitations, and future work.

These objectives reflect the real nature of the project. The system does not predict predefined topic labels. Instead, it discovers topic structures from unlabeled comments and evaluates the usefulness of these discovered structures through downstream BERTopic behavior and manual interpretation.

1.4 RESEARCH CONTRIBUTIONS

This project provides several technical and methodological contributions related to unsupervised topic modeling of news comments.

First, the project presents a complete topic analysis pipeline for news comments. The pipeline begins with raw comments, applies cleaning and filtering, preserves a post-level split, mines semantic pairs, fine-tunes an embedding model, generates comment embeddings, applies BERTopic, and evaluates the discovered topics.

Second, the project uses a post-level split strategy to reduce data leakage. Instead of randomly splitting comments, all comments belonging to the same news post remain inside the same split. This is important because comments under the same post often share a common discussion context.

Third, the project constructs training data without manually labeled topic classes. Since no gold topic labels are available, the system uses automatically mined **Comment** ↔ **Comment** positive pairs. These pairs are selected from comments belonging to the same post and filtered using cosine similarity thresholds.

Fourth, the project fine-tunes a SentenceTransformer model specifically for news comments. The final adopted model, named the **Final Fine-Tuned MiniLM Comment Encoder**, was trained on **1,500,000** positive training pairs and **30,000** validation pairs using **MultipleNegativesRankingLoss**.

Fifth, the project separates embedding-level evaluation from downstream topic modeling evaluation. Training loss, retrieval metrics, and cosine sanity checks are used to understand the behavior of the encoder, but the final model decision is based on downstream BERTopic performance because the project goal is topic modeling, not retrieval optimization.

Sixth, the project compares the final model with previous and additional models. This includes V5 Stage 1 Controlled, V6-B, V8, and V9 BGE. These comparisons show that the final model was not selected merely because it was the latest experiment, but because it achieved the strongest downstream BERTopic balance on the fixed evaluation pool.

Finally, the project includes manual topic inspection to identify strong, mixed, and weak topics. This improves the reliability of interpretation because unsupervised topic modeling cannot be fully judged by numerical metrics alone.

1.5 SCOPE AND BENEFICIARIES

The scope of this project is limited to unsupervised topic modeling of news comments. The system does not perform sentiment analysis, supervised classification, fact-checking, user profiling, political stance detection, or automatic truth verification. It also does not assign human-defined topic labels to comments. Instead, it discovers topics based on semantic similarity and clustering.

The project focuses on text cleaning, post-level data splitting, comment-comment pair mining, SentenceTransformer fine-tuning, embedding generation, BERTopic topic extraction, topic modeling evaluation, and manual topic inspection.

The system can benefit researchers, media analysts, public opinion analysts, data science teams, academic supervisors, and decision makers who need to explore large collections of news comments. It can help summarize public discussions, identify recurring themes, and analyze noisy comment datasets in a structured way.

1.6 REPORT ORGANIZATION

This report is organized into six chapters.

Chapter 1 introduces the research context, scientific problem, objectives, contributions, scope, beneficiaries, and report organization.

Chapter 2 presents the theoretical study. It explains topic modeling, unsupervised learning in text analysis, short-text challenges, classical topic modeling, transformer-based embeddings, SentenceTransformer, MiniLM, cosine similarity, contrastive fine-tuning, UMAP, HDBSCAN, BERTopic, c-TF-IDF, and the evaluation metrics used in the project.

Chapter 3 presents the literature review. It summarizes recent related studies, compares modern techniques and systems, and identifies the research gap addressed by the proposed solution.

Chapter 4 explains the research methodology. It describes the dataset, post-level split, cleaning process, pair mining strategy, embedding model training, model development history, final BERTopic configuration, and implementation environment.

Chapter 5 presents the research results. It explains the evaluation strategy, reports embedding-level results, presents final BERTopic results, compares the proposed and non-adopted models, analyzes confidence and topic diversity, discusses rejected models, and provides manual topic inspection.

Chapter 6 concludes the report by summarizing the main findings, discussing challenges, and presenting future work directions.

CHAPTER 2

THEORETICAL STUDY

This chapter presents the theoretical foundations of the project, including topic modeling, unsupervised text analysis, short-text challenges, transformer-based embeddings, SentenceTransformer, MiniLM, contrastive fine-tuning, UMAP, HDBSCAN, BERTopic, c-TF-IDF, and the evaluation metrics used in this work.

2.1 TOPIC MODELING

Topic modeling is a text analysis technique used to discover hidden thematic structures inside a collection of documents. Instead of reading each document manually, a topic modeling method attempts to group documents according to recurring semantic patterns. In this project, the documents are news comments, and the objective is to discover the main discussion themes that appear across a large collection of user-generated comments.

A topic can be understood as a group of semantically related comments represented by a set of important words. For example, a topic may contain representative words such as **russia**, **putin**, and **russian**, which indicates a geopolitical discussion. Another topic may contain words such as **healthcare**, **insurance**, and **obamacare**, which indicates a health-policy discussion. The role of topic modeling is to discover these structures automatically from the data.

Topic modeling differs from supervised classification. In supervised classification, each document has a known label, and the model learns to predict that label. In topic modeling, the labels are not known in advance. The system attempts to discover latent topics from the documents themselves. Therefore, the output is exploratory rather than a direct prediction of predefined classes.

2.2 UNSUPERVISED LEARNING IN TEXT ANALYSIS

The current project follows an unsupervised topic modeling approach because the dataset does not contain manually annotated topic labels. This means that comments are not labeled as politics, health, religion, economy, or any other fixed class. The system must infer useful structures from the data without relying on a gold-standard topic label for each comment.

In unsupervised text analysis, evaluation is more difficult than in supervised learning. There is no direct ground-truth label that can be used to calculate final accuracy, precision, recall, or F1-score. Instead, the quality of the discovered topics must be assessed using indicators such as the number of comments assigned to valid topics, the noise ratio, confidence scores, topic diversity, and manual inspection of the produced topics.

The project is therefore not a supervised classification system. It is an unsupervised topic modeling system that uses fine-tuned semantic embeddings to improve the quality of clustering and topic extraction.

2.3 SHORT TEXT AND NEWS COMMENT CHALLENGES

News comments are challenging because they are short, informal, noisy, and highly context-dependent. A formal article usually contains a clear structure, multiple paragraphs, repeated key terms, and enough context for statistical text analysis. In contrast, a news comment may consist of only a few words, a reaction, a joke, a name, a quotation, or a reply to another user. This makes automatic topic discovery more difficult.

Short comments provide limited lexical evidence. A comment may express a strong opinion without explicitly mentioning the main subject of the news post. Other comments may include sarcasm, spelling variation, emotional language, repeated phrases, or unrelated side discussions. These properties increase noise and make it harder for a topic modeling algorithm to form coherent clusters.

Another difficulty is that comments under the same news post often share an implicit context. For example, users may discuss a political event without repeating the names of the people or countries involved. Therefore, a purely word-based method may fail to recognize that several short comments refer to the same issue. This motivates the use of semantic embeddings, which can represent meaning beyond exact word overlap.

2.4 CLASSICAL TOPIC MODELING

Classical topic modeling methods such as **Latent Dirichlet Allocation** and **Non-negative Matrix Factorization** are widely used in text analysis. LDA assumes that documents are mixtures of topics and that topics are distributions over words. NMF decomposes a document-term matrix into lower-dimensional components that can be interpreted as topics.

These methods can be useful when documents are long enough to provide stable word co-occurrence patterns. However, they often struggle with short and noisy texts because such texts contain limited vocabulary and weak co-occurrence evidence. They also depend heavily on lexical overlap. Two comments may discuss the same issue using different words, but a purely word-based method may not recognize their semantic similarity.

This project does not claim to experimentally outperform LDA or NMF unless such experiments are explicitly conducted and verified. Instead, classical methods are discussed as theoretical background. The implemented system adopts an embedding-based approach because semantic embeddings are more suitable for representing short, noisy, and informal news comments.

2.5 TRANSFORMER-BASED SENTENCE EMBEDDINGS

Transformer-based models are neural architectures designed to model contextual relationships between tokens in text. Unlike older sequential architectures, transformers use attention mechanisms to capture dependencies between words regardless of their position in the sequence. This makes them effective for representing complex linguistic meaning.

In natural language processing, transformer models are used for tasks such as classification, translation, summarization, retrieval, semantic similarity, and clustering. However, a standard transformer model is not always optimized to produce sentence embeddings suitable for direct similarity comparison. Sentence embedding models address this limitation by producing fixed-size vectors that represent the semantic meaning of sentences or documents.

In this project, each cleaned news comment is converted into a dense numerical vector called an embedding. The quality of this embedding directly affects the performance of the topic modeling pipeline. If semantically related comments are close in the embedding space, dimensionality reduction and clustering are more likely to produce coherent topic groups. If the embedding space is weak, clustering may produce fragmented topics or a high number of noise comments.

Transformer-based embeddings are appropriate for this project because news comments may express similar meanings with different words. For example, comments about healthcare may mention hospitals, insurance, medicine, policy, or treatment. A semantic embedding model can represent these comments as related even when exact word overlap is limited.

2.6 SENTENCETRANSFORMER AND MINILM

SentenceTransformer is a framework for generating sentence-level and document-level embeddings. It is commonly used in semantic search, clustering, paraphrase mining, retrieval, and similarity comparison. In this project, SentenceTransformer is used to encode news comments into semantic vectors before applying the BERTopic pipeline.

The base model used in this project is **sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2**. This model produces compact multilingual embeddings with a vector dimension of 384. It is efficient enough for large-scale embedding generation while still providing useful semantic representation. However, the base model is general-purpose and was not specifically trained on the project's news comment dataset.

To adapt the base MiniLM model to the project domain, the model was fine-tuned using automatically mined **Comment** ↔ **Comment** positive pairs. The final model is academically named the **Final Fine-Tuned MiniLM Comment Encoder**, while its internal experiment identifier is **V7**. The purpose of fine-tuning is not to teach the model predefined topic labels, but to improve the semantic geometry of the embedding space.

A better embedding space should place semantically related comments closer together and unrelated comments farther apart. This property is important because BERTopic depends on embeddings before applying UMAP and HDBSCAN. If the embedding model captures the relationships between comments more effectively, the downstream topic modeling stage can assign more comments to valid topics and reduce the number of noise comments.

2.7 COSINE SIMILARITY

Cosine similarity is a common measure used to compare two embedding vectors. It measures the cosine of the angle between two vectors rather than their absolute magnitude. This makes it suitable for comparing normalized semantic embeddings.

Equation 2.1: Cosine Similarity

$$\cos(A, B) = (A \cdot B) / (\|A\| \|B\|)$$

In this equation, **A** and **B** are embedding vectors, **A · B** is the dot product between the vectors, and **||A||** and **||B||** are their vector norms. A cosine similarity close to 1 indicates high semantic similarity, while a value near 0 indicates weak similarity.

In this project, cosine similarity is used in two important stages. First, it is used during pair mining to filter candidate comment pairs. Second, it is used after training as a sanity check to compare the average similarity of positive pairs with the average similarity of random pairs. This helps verify that the embedding model separates related comments from unrelated comments.

During pair mining, a candidate pair is accepted only if its similarity falls within a controlled range. The lower threshold removes weak semantic pairs, while the upper threshold removes near-duplicates or overly easy pairs. This improves the quality of the training data used for contrastive fine-tuning.

2.8 CONTRASTIVE FINE-TUNING AND MULTIPLENEGATIVESRANKINGLOSS

Contrastive fine-tuning is a training approach that improves an embedding model by teaching it to place related examples close together and unrelated examples farther apart. In the context of this project, each positive training example consists of two comments that are expected to be semantically related. These examples are represented as **Comment** $\leftrightarrow$ **Comment** positive pairs.

The project uses this approach because manually labeled topic classes are not available. The model is not trained as a supervised classifier. Instead, it is trained to improve semantic similarity between related comments. This is more suitable for a topic modeling system because the final objective is not to predict a predefined class, but to produce embeddings that support better clustering.

The training objective used for the final model is **MultipleNegativesRankingLoss**. This loss is commonly used in SentenceTransformer training when positive pairs are available. For each positive pair in a batch, the paired comment is treated as the correct match, while other comments in the same batch act as implicit negative examples.

Equation 2.2: Conceptual Objective of MultipleNegativesRankingLoss

The model maximizes the similarity of each positive pair while minimizing its similarity to other in-batch examples that are treated as negatives.

This training strategy is efficient because it does not require explicitly labeled negative pairs. The batch itself provides negative examples. A larger batch can provide more in-batch negatives, which may improve the contrastive learning signal. In the final model, the training batch contains positive comment pairs, and each pair contributes to shaping the semantic embedding space.

The role of contrastive fine-tuning in this project is methodological. It improves the encoder before the BERTopic pipeline is applied. However, embedding-level metrics such as training loss or retrieval accuracy are not sufficient to select the final model. The final decision must be based on downstream topic modeling quality, because the project goal is topic discovery.

2.9 UMAP

UMAP, or Uniform Manifold Approximation and Projection, is a dimensionality reduction algorithm used to project high-dimensional data into a lower-dimensional space. In this project, the input to UMAP is the set of semantic embeddings generated by the final fine-tuned SentenceTransformer model.

Sentence embeddings are high-dimensional vectors. The final MiniLM-based model produces embeddings with a dimension of 384. Direct clustering in such a high-dimensional space can be difficult because distance relationships become harder to interpret and density estimation becomes less stable. UMAP helps by reducing the embeddings to a smaller number of dimensions while preserving important neighborhood relationships.

UMAP is not the topic model itself. It is an intermediate step that prepares the embedding space for clustering. After UMAP reduces the dimensionality of the embeddings, HDBSCAN is applied to identify dense regions that correspond to potential topic clusters. Therefore, UMAP strongly affects the downstream clustering behavior.

The final BERTopic configuration used the following UMAP parameters:

- **n_neighbors = 15**: controls the balance between local and global structure.
- **n_components = 5**: reduces embeddings to five dimensions before clustering.
- **min_dist = 0.0**: allows points to be packed closely in the reduced space.
- **metric = cosine**: uses cosine distance, which is appropriate for semantic embeddings.
- **random_state = 42**: supports reproducibility of the dimensionality reduction process.

These parameters were selected to support stable topic extraction. A low number of components makes the data more suitable for density-based clustering, while cosine-based neighborhood preservation aligns with the way semantic embeddings are usually compared.

In the final pipeline, UMAP acts as a bridge between the embedding model and HDBSCAN. It does not generate topics directly, but it reshapes the embedding space so that topic clusters can be detected more effectively.

2.10 HDBSCAN

HDBSCAN, or Hierarchical Density-Based Spatial Clustering of Applications with Noise, is a clustering algorithm that detects dense regions in data. Unlike algorithms that require a predefined number of clusters, HDBSCAN can discover the number of clusters automatically based on density. This is suitable for news comments because the number of discussion topics is not known in advance.

One important property of HDBSCAN is its ability to assign uncertain or isolated points to noise. In BERTopic, the noise label is usually represented as **Topic -1**. Comments assigned to this label are not considered part of a valid topic. This behavior is useful because not every news comment belongs to a coherent discussion theme. Some comments are too short, generic, ambiguous, or unrelated.

In this project, noise is not treated as a failure by itself. Instead, it is an expected part of density-based topic modeling. However, a lower noise ratio is generally preferred when it is achieved without producing weak or fragmented topics. Therefore, the final model is evaluated by considering noise ratio together with valid topic coverage, confidence, detected topic count, topic diversity, and manual topic inspection.

The final HDBSCAN configuration used the following parameters:

- **min_cluster_size = 100**: requires a topic cluster to contain at least 100 comments.
- **min_samples = 10**: controls how conservative the clustering is when identifying dense regions.
- **metric = euclidean**: used after UMAP dimensionality reduction.
- **cluster_selection_method = eom**: selects clusters using the excess of mass method.
- **prediction_data = True**: enables additional prediction-related functionality in HDBSCAN.

The value of **min_cluster_size** is especially important. A very small value may produce many fragmented topics, while a very large value may merge distinct topics or increase noise. The selected value supports a more compact and controlled topic structure.

2.11 BERTOPIC AND C-TF-IDF

BERTopic is an embedding-based topic modeling framework. Its general pipeline consists of generating document embeddings, reducing dimensionality, clustering documents, and extracting topic representations. This makes it appropriate for the current project because the quality of the topic model can be improved by improving the semantic embeddings used as input.

In this project, embeddings are generated separately using the final fine-tuned SentenceTransformer model. Therefore, BERTopic is configured with **embedding_model = None**. This is intentional. It means that BERTopic does not internally generate embeddings; instead, precomputed embeddings are passed directly to the fitting process.

The simplified BERTopic pipeline used in the project is:

Cleaned Comments → Precomputed Embeddings → UMAP → HDBSCAN → c-TF-IDF Topic Representation

After clustering, BERTopic uses **c-TF-IDF**, or class-based Term Frequency-Inverse Document Frequency, to extract representative words for each topic. Instead of calculating term importance for individual documents, c-TF-IDF treats each topic as a class and identifies words that are important for that topic compared with other topics.

This representation step is essential because clustering alone produces numerical topic IDs, which are not directly interpretable. c-TF-IDF helps convert each cluster into a set of representative words. These words allow the researcher to inspect and interpret the discovered topic. For example, a topic represented by words such as **vote**, **voter**, **fraud**, and **election** can be interpreted as an election-related topic.

BERTopic is therefore not a single algorithm but a pipeline that combines semantic representation, dimensionality reduction, clustering, and topic-word extraction. The performance of the final system depends on the interaction between these components.

2.12 EVALUATION METRICS AND EQUATIONS

Because this project is an unsupervised topic modeling project, final evaluation does not use classification accuracy, precision, recall, or F1-score as primary metrics. These metrics require ground-truth labels, while the dataset used in this project does not contain manually assigned topic classes. Instead, the project uses topic modeling metrics that describe the behavior of BERTopic output.

2.12.1 Noise Ratio

Noise Ratio measures the percentage of comments assigned to the HDBSCAN noise label -1. A lower noise ratio is generally better because it means that more comments were assigned to valid topic clusters.

Equation 2.3: Noise Ratio

$$\text{Noise Ratio} = (\text{Noise Comments} / \text{Total Documents}) \times 100$$

2.12.2 Valid Ratio

Valid Ratio measures the percentage of comments assigned to valid topic clusters rather than noise. It is the complement of the noise ratio when every document is assigned either to a valid topic or to noise.

Equation 2.4: Valid Ratio

$$\text{Valid Ratio} = (\text{Valid Topic Comments} / \text{Total Documents}) \times 100$$

2.12.3 Valid Topic Comments

Valid Topic Comments counts the number of comments that were assigned to actual topics. This metric is important because the project aims to organize as many meaningful comments as possible into interpretable topic clusters.

Equation 2.5: Valid Topic Comments

Valid Topic Comments = Total Documents - Noise Comments

2.12.4 Detected Topics

Detected Topics measures the number of valid topic clusters discovered by BERTopic, excluding the noise label. A smaller number is not always better, and a larger number is not always better. The value must be interpreted together with topic quality, noise ratio, confidence, and manual inspection.

Equation 2.6: Detected Topics

Detected Topics = Number of unique topic labels excluding -1

2.12.5 Average Confidence

Average Confidence measures the average confidence score assigned by the topic modeling process. In this project, confidence is interpreted in two forms. Average Confidence All is calculated over all comments, including noise. Average Confidence Valid is calculated only over comments assigned to valid topics.

Equation 2.7: Average Confidence All

Average Confidence All = $\sum \text{confidence}_i / N$

Equation 2.8: Average Confidence Valid

Average Confidence Valid = $\sum \text{confidence}_i \text{ for topic}_i \neq -1 / \text{Number of Valid Topic Comments}$

2.12.6 Median Confidence Valid

Median Confidence Valid calculates the median confidence value among comments assigned to valid topics. The median is useful because it is less sensitive to extreme values than the mean.

Equation 2.9: Median Confidence Valid

Median Confidence Valid = $\text{median}(\text{confidence}_i \text{ for topic}_i \neq -1)$

2.12.7 Topic Diversity

Topic Diversity measures the diversity of representative topic words. It is calculated by dividing the number of unique words among the top topic words by the total number of top words considered. In this project, the top words are taken from the topic representations generated by c-TF-IDF.

Equation 2.10: Topic Diversity

Topic Diversity = $\text{Number of Unique Words among Top-N Topic Words} / \text{Total Number of Top-N Topic Words}$

Topic Diversity is useful, but it should not be interpreted alone. A model may achieve a slightly higher diversity score while producing more noise or fewer valid topic comments. Therefore, final model selection must consider the full downstream BERTopic balance, including coverage, noise, confidence, detected topics, diversity, and manual inspection.

2.13 CHAPTER SUMMARY

This chapter presented the theoretical foundations of the project. It explained why the project is an unsupervised topic modeling task, why short news comments are difficult to analyze, and why embedding-based topic modeling is more suitable than purely word-based approaches. It also introduced SentenceTransformer, MiniLM, cosine similarity, contrastive fine-tuning, MultipleNegativesRankingLoss, UMAP, HDBSCAN, BERTopic, c-TF-IDF, and the evaluation metrics used in the project.

The next chapter reviews recent related studies and identifies the research gap addressed by the proposed system.

CHAPTER 3

LITERATURE REVIEW

This chapter reviews recent studies and modern approaches related to topic modeling, short-text analysis, transformer-based embeddings, and BERTopic-based systems. It also identifies the research gap addressed by the proposed news comment topic analysis system.

3.1 INTRODUCTION

The purpose of this chapter is to review previous work related to the problem addressed in this project. Since the project focuses on unsupervised topic modeling of news comments, the literature review concentrates on four main areas: classical topic modeling, short-text and social media topic modeling, embedding-based topic modeling, and transformer-based semantic representation.

The reviewed studies show that topic modeling has moved from purely word-frequency-based approaches toward semantic and embedding-based approaches. This transition is important because short user-generated texts, such as tweets, comments, and online discussions, do not usually contain enough lexical context for classical word co-occurrence methods to work reliably. For this reason, recent research increasingly uses sentence embeddings, transformer-based models, clustering, and topic representation techniques.

This project is directly related to that research direction. It does not attempt to build a supervised classifier. Instead, it builds an unsupervised topic modeling pipeline that improves the embedding space before applying BERTopic. Therefore, the most relevant studies are those that address short-text topic modeling, BERTopic, semantic clustering, transformer embeddings, and topic interpretation.

The chapter is organized as follows. Section 3.2 reviews recent studies in topic modeling and short-text analysis. Section 3.3 presents a comparative table of related work. Section 3.4 discusses modern techniques and frameworks used in this domain. Section 3.5 compares major system-level approaches. Section 3.6 identifies the research gap and explains how the proposed system addresses it.

The literature review is used to position the current project academically. It does not claim direct numerical superiority over previous studies unless the datasets, evaluation protocols, and experimental conditions are equivalent.

3.2 RECENT STUDIES IN TOPIC MODELING

Grootendorst introduced BERTopic as an embedding-based topic modeling framework that combines transformer embeddings, dimensionality reduction, clustering, and class-based TF-IDF topic representation [1]. The importance of BERTopic lies in its modular pipeline. It allows the researcher to use external embedding models, reduce the dimensionality of document vectors, cluster documents, and then extract interpretable topic words. This makes BERTopic highly relevant to the current project.

Egger and Yu compared several topic modeling methods, including LDA, NMF, Top2Vec, and BERTopic, on short social media posts [2]. Their study is relevant because social media posts share several properties with news comments: they are short, informal, noisy, and context-dependent. Their findings support the idea that embedding-based methods can be more suitable for short and noisy documents than classical word-frequency-based methods.

Doogan, Buntine, and Linger reviewed the use of topic models for short-text social media analysis [3]. Their work highlights the methodological difficulty of extracting stable topics from short documents. Short texts often contain limited vocabulary, weak co-occurrence signals, and high ambiguity. This supports the main assumption of the current project: improving semantic representation before clustering can help produce more coherent topics.

Pham and colleagues proposed TopicGPT as a framework for using large language models in topic interpretation and topic-related analysis [4]. This work is relevant to the current project as a possible future direction. The proposed system does not use an LLM for clustering, but LLM-based topic labeling may be considered later as an auxiliary interpretation layer.

3.2 RECENT STUDIES IN TOPIC MODELING (CONTINUED)

Hellwig and colleagues applied BERTopic to Twitter discourse analysis [5]. This study is highly relevant because Twitter discourse and news comment sections share several practical challenges: short texts, informal language, public opinion expression, and noisy user interaction. Their results demonstrate the effectiveness of combining density-based clustering with semantic representations for discovering hidden themes in public discourse.

Across these studies, a consistent observation appears: short and noisy user-generated text requires careful preprocessing, strong semantic representation, and cautious evaluation. Numerical metrics alone are not sufficient. Manual inspection remains important because unsupervised topics may be coherent, mixed, generic, or weak depending on the structure of the data and the behavior of the clustering algorithm.

The current project builds directly on this research direction by using a fine-tuned SentenceTransformer encoder before applying the BERTopic pipeline. The primary goal is not merely to apply an existing framework, but to structurally improve the embedding space on which the clustering step relies.

3.3 COMPARATIVE TABLE OF RELATED STUDIES

Table 3.1 summarizes the five main related studies and explains their relevance to the current project. The comparison focuses on the method used, the main strength, the limitation, and the relation to the proposed system.

Table 3.1: Summary of Recent Related Studies

Study	Year	Main Method	Strength	Limitation	Relevance to Project
Grootendorst [1]	2022	BERTopic	Combines embeddings, clustering, and c-TF-IDF.	Topic quality depends heavily on embeddings.	Core framework used in the proposed system.
Egger & Yu [2]	2022	LDA, BERTopic	Compares classical and embedding methods.	Results depend on dataset type.	Supports embedding methods for short text.
Doogan et al. [3]	2023	Short-text review	Explains difficulties of noisy social texts.	Review-oriented only.	Supports the need for strong semantic vectors.
Pham et al. [4]	2024	TopicGPT	Uses LLMs for topic interpretation.	High cost and dependency on external APIs.	Relevant for future topic labeling extensions.
Hellwig et al. [5]	2024	BERTopic for discourse	Applies BERTopic to social media discourse.	Requires manual interpretation of clusters.	News comments share traits with Twitter data.

The reviewed studies indicate that most topic modeling pipelines depend heavily on the quality of the representation used before clustering. Therefore, the current project focuses on fine-tuning the embedding model itself to adapt to the comment domain.

This comparative review supports the project's methodological direction: use embedding-based topic modeling, improve the embedding space via fine-tuning, evaluate downstream clustering quality, and perform manual topic inspection.

3.4 MODERN TECHNIQUES AND FRAMEWORKS

Recent topic modeling systems increasingly depend on semantic representations rather than only word frequencies. In this direction, transformer-based embeddings have become important because they represent the meaning of a text as a dense vector. These vectors can then be used for semantic search, clustering, retrieval, and topic modeling.

SentenceTransformer is one of the most relevant frameworks for this project. It provides models that generate sentence-level or document-level embeddings suitable for similarity comparison. Unlike raw transformer outputs, SentenceTransformer models are designed to produce embeddings that can be compared directly using measures such as cosine similarity. This makes them suitable for clustering news comments.

BERTopic is another central framework. It is not only a topic representation method; it is a complete modular pipeline. The pipeline can accept precomputed embeddings, apply UMAP for dimensionality reduction, use HDBSCAN for clustering, and then extract topic words using c-TF-IDF. This modular design allows the current project to replace the default embedding stage with a fine-tuned embedding model.

UMAP and HDBSCAN are also important. UMAP reduces high-dimensional embeddings to a lower dimensional space that is more suitable for clustering. HDBSCAN identifies dense clusters and assigns uncertain comments to noise. This is appropriate for noisy news comments because not every comment belongs to a meaningful topic.

Large language models represent another modern direction in topic modeling. They can be useful for topic labeling, summarization, and interpretation. However, the current project does not depend on an LLM to generate topics. The main system remains an embedding-based unsupervised topic modeling pipeline. LLM-based interpretation can be considered as a future enhancement rather than a core component of the final model.

3.5 COMPARATIVE ANALYSIS OF EXISTING APPROACHES

Several approaches can be used to analyze topics in large text collections. Each approach has strengths and limitations. Table 3.2 compares major approach families and explains how they relate to the proposed system.

Table 3.2: Comparative Analysis of Topic Modeling Approaches

Approach	Example	Strength	Weakness	Relation to This Project
Classical topic modeling	LDA, NMF	Interpretable and widely used for structured documents.	Weak with short, noisy, and low-context comments.	Used as theoretical background, not as the final implemented approach.
Embedding-based topic modeling	BERTopic	Captures semantic similarity beyond exact word overlap.	Highly dependent on embedding quality and clustering configuration.	Main approach used in the proposed system.
Semantic clustering	Embedding + UMAP + HDBSCAN	Can discover topics without predefined labels.	Requires careful evaluation and manual inspection.	Forms the downstream structure of the final pipeline.
LLM-assisted topic interpretation	TopicGPT-like methods	Can help label or summarize discovered topics.	May introduce cost, dependency, and reproducibility concerns.	Possible future extension, not the core method.

The comparison shows that no single approach solves all aspects of the problem. Classical methods are simple and interpretable but struggle with noisy short comments. LLM-assisted methods can support interpretation but may introduce operational and reproducibility concerns. Embedding-based topic modeling offers the best fit for this project because it can discover topics without labeled data while still representing semantic meaning.

3.6 RESEARCH GAP AND PROPOSED INNOVATION

The literature review reveals several gaps that are directly related to the current project. First, short and noisy user-generated texts remain difficult for topic modeling systems. Many comments contain limited context, informal expressions, sarcasm, emotional reactions, or unrelated side discussions. This weakens the performance of approaches that rely mainly on word co-occurrence.

Second, embedding-based topic modeling methods depend heavily on the quality of the embedding model. Applying BERTopic with a general-purpose embedding model may produce useful results, but the embedding model may not be optimized for the specific semantic patterns of news comments. This creates a need to adapt the embedding space to the target domain before clustering.

Third, many topic modeling workflows focus only on the final topic model and do not clearly separate embedding-level evaluation from downstream topic-model evaluation. This separation is important. A model may achieve lower training loss or stronger retrieval metrics, but still produce worse topic clusters after UMAP, HDBSCAN, and BERTopic. The current project explicitly separates these two evaluation levels.

Fourth, random comment-level splitting can introduce data leakage when comments from the same news post appear in different splits. Since comments under the same post often share contextual meaning, such leakage may produce overly optimistic evaluation. The current project addresses this issue using a post-level split strategy.

The proposed innovation is the combination of post-level split, automatically mined Comment $\leftrightarrow$ Comment positive pairs, fine-tuned MiniLM-based sentence embeddings, and downstream BERTopic evaluation on a fixed test pool.

The project therefore contributes a practical and methodologically controlled pipeline for unsupervised topic modeling of news comments. The final adopted model is not selected based only on training loss. It is selected because it produces the best downstream BERTopic balance in terms of valid topic coverage, noise reduction, confidence, compact topic structure, and stable topic diversity.

3.7 CHAPTER SUMMARY

This chapter reviewed recent studies and modern approaches related to the proposed news comment topic analysis system. The reviewed literature shows that topic modeling has moved from classical word-frequency-based methods toward semantic and embedding-based approaches. This transition is especially important for short and noisy texts, where lexical overlap alone is often insufficient.

BERTopic is highly relevant to the current project because it combines semantic embeddings, dimensionality reduction, clustering, and c-TF-IDF topic representation. However, the effectiveness of BERTopic depends strongly on the quality of the embeddings used as input. This supports the decision to fine-tune a SentenceTransformer model before applying the BERTopic pipeline.

The review also shows that short-text topic modeling requires careful interpretation. Numerical metrics alone are not enough to judge topic quality. For this reason, the proposed system combines quantitative evaluation with manual topic inspection. This helps identify strong topics, weak topics, mixed topics, and generic clusters.

The main research gap identified in this chapter is the need for a controlled pipeline that improves the semantic representation of news comments before clustering, prevents data leakage through post-level splitting, and evaluates the final result at the downstream topic-modeling level. The proposed system addresses this gap through Comment $\leftrightarrow$ Comment pair mining, contrastive fine-tuning of a MiniLM-based SentenceTransformer model, and final evaluation using BERTopic metrics on a fixed test pool.

The next chapter presents the research methodology in detail, including dataset preparation, post-level split, text cleaning, pair mining, embedding model training, model development history, final BERTopic configuration, and implementation environment.

CHAPTER 4

RESEARCH METHODOLOGY

This chapter presents the complete methodology of the proposed system, including the dataset, post-level split, text cleaning, pair mining, embedding model training, model development history, final BERTopic configuration, and implementation environment.

4.1 INTRODUCTION

This chapter explains the methodology used to build the proposed news comment topic analysis system. The system is designed as an unsupervised topic modeling pipeline, which means that it does not depend on manually labeled topic categories. Instead, it improves the semantic representation of comments and then applies clustering-based topic modeling to discover latent discussion themes.

The methodology contains two connected levels. The first level is the embedding-training level, where a SentenceTransformer model is fine-tuned using automatically mined Comment $\leftrightarrow$ Comment positive pairs. The second level is the downstream topic-modeling level, where the generated embeddings are passed to BERTopic through UMAP, HDBSCAN, and c-TF-IDF.

This separation is necessary because improving an embedding model during training does not automatically guarantee better topic modeling results. Therefore, the methodology evaluates the embedding model using training-side indicators, but the final model selection is based on downstream BERTopic performance.

4.2 GENERAL SYSTEM PIPELINE

The complete system follows a sequential pipeline. Each stage prepares the output needed by the next stage. The general pipeline is shown below.

**Raw News Comments $\rightarrow$ Text Cleaning $\rightarrow$ Post-Level Split $\rightarrow$ Pair Mining
 $\rightarrow$ Fine-Tuned SentenceTransformer $\rightarrow$ Comment Embeddings
 $\rightarrow$ UMAP $\rightarrow$ HDBSCAN $\rightarrow$ BERTopic Topics $\rightarrow$ Metrics & Manual Inspection**

The final adopted embedding model is the Final Fine-Tuned MiniLM Comment Encoder, internally referred to as V7. Its role is not to classify comments into predefined categories. Instead, it produces semantic embeddings that help BERTopic discover topic clusters from unlabeled news comments.

4.3 DATASET DESCRIPTION

The project uses a large-scale news comment dataset. The raw comments file contains more than one million records. The main textual field is the comment message, while the other fields provide metadata related to the comment author and source post.

Table 4.1: Raw Comment Dataset Structure

Column	Description
created_time	Date and time when the comment was created.
from_id	Identifier of the comment author. It is metadata and not a topic label.
from_name	Name of the comment author. It is not used as a supervised label.
message	Raw comment text. This is the main textual field used for modeling.
post_name	Identifier or composite reference related to the source news post.

The raw comments dataset has the following shape:

```
Raw comments shape = (1,038,319, 5)
```

The **message** column is the primary source of textual content. User-related fields such as **from_id** and **from_name** are not used as labels. They do not define the topic of a comment and are not treated as classification targets.

The project also uses post-level information to organize comments according to their source news posts. This is important because comments under the same post often share contextual meaning. For example, many users may respond to the same political, economic, health-related, or social event. This post-level relationship is used to prevent data leakage and to support pair mining.

4.4 POST-LEVEL SPLIT AND DATA LEAKAGE PREVENTION

A central methodological decision in this project is the use of a post-level split. Instead of randomly splitting individual comments into training, validation, and test sets, the project keeps all comments belonging to the same news post inside the same split.

This strategy is necessary because random comment-level splitting may cause data leakage. If comments from the same news post appear in both training and testing sets, the model may indirectly learn discussion-specific patterns from the training data. This can make the evaluation results appear stronger than they actually are.

Post-level splitting ensures that the model is evaluated on comments from posts that were not used during training or pair mining.

The post split file has the following structure:

Post split shape = (17,351, 4)

Table 4.2: Post Split Structure

Column	Description
post_id_true	Unique post identifier used to group comments.
comment_count	Number of comments associated with the post.
page_source	Source page or origin of the post.
split	Assigned split: train, validation, or test.

Data leakage prevention is especially important because pair mining uses comments from the same post. If post boundaries were not respected, semantically related comments from the same discussion could appear across different splits. Therefore, the post-level split is a fundamental part of the methodology, not a minor preprocessing detail.

4.5 TEXT CLEANING AND CLEAN POOLS

Text cleaning was applied to reduce technical noise while preserving useful semantic information. Since the project depends on semantic embeddings, the cleaning strategy avoids overly aggressive transformations that may remove important context.

The cleaning process focuses on preparing comments for embedding generation and topic modeling. It includes removing or normalizing URLs, handling HTML artifacts, normalizing whitespace, and ensuring that the text field is usable for model inference.

Table 4.3: Filtering Summary

Stage	Count
Before filtering	1,035,495
After filtering	843,636
Removed comments	191,859

After linking comments to the post-level split and filtering low-value entries, the filtered comments were distributed across training, validation, and test splits.

Table 4.4: Filtered Comment Split Distribution

Split	Number of Comments	Purpose
Train	593,543	Used for pair mining and model training.
Validation	125,614	Used for validation pair mining and training evaluation.
Test	124,479	Reserved from training and pair mining.

The final BERTopic evaluation used a fixed clean test pool containing **58,014** comments. Using a fixed evaluation pool enables fair comparison between models because differences in results are caused by the model and configuration, not by changes in the evaluation data.

4.6 PAIR MINING STRATEGY

Because the project does not have manually labeled topic classes, the embedding model could not be trained as a supervised classifier. Instead, the project uses automatically mined positive pairs. The pair type used in the final model is:

Comment ↔ Comment positive pairs

Each training example consists of two comments that are expected to be semantically related. These pairs are selected from comments belonging to the same news post and then filtered using cosine similarity. The assumption is that comments under the same post often discuss the same event or issue, but not every pair from the same post is useful.

4.6.1 Why Pair Mining Was Needed

The goal of pair mining is to construct training data without manual topic labels. Since the system is designed for unsupervised topic modeling, it does not require a human to label each comment with a predefined topic. Instead, it uses weak semantic signals from news discussions to construct positive pairs for contrastive fine-tuning.

4.6.2 Comment-to-Comment Pair Construction

Candidate pairs are generated from comments inside the same post. However, to avoid weak pairs and trivial duplicates, each candidate pair is filtered using cosine similarity. This helps select pairs that are semantically related but not identical.

4.6.3 Similarity Thresholds

The final pair mining process uses a controlled similarity range:

$0.45 \leq \text{cosine similarity} \leq 0.90$

The lower threshold removes weak or unrelated pairs. The upper threshold removes near-duplicates and overly easy pairs. This range supports contrastive training by keeping pairs that are related enough to be useful but not so similar that they fail to provide a meaningful learning signal.

4.6 PAIR MINING STRATEGY (CONTINUED)

4.6.4 Final Pair Mining Statistics

The final pair mining setup used large-scale mined Comment $\leftrightarrow$ Comment pairs. These pairs are not supervised class labels. They are semantic training pairs used to improve the embedding encoder.

Table 4.5: Final Pair Mining Results

Item	Training	Validation	Meaning
Comment pool	593,543	125,614	Comments used for pair mining.
Similarity threshold	0.45 to 0.90	0.45 to 0.90	Rejects weak pairs and near-duplicates.
Candidate pairs after filtering and deduplication	3,701,173	752,789	Available pairs before final sampling.
Final selected pairs	1,500,000	30,000	Final training and validation setup.
Mean pair similarity	0.6024	0.6075	Pairs are semantically related but not trivial duplicates.
Max comments per post	200	200	Prevents very large posts from dominating pair construction.

The mean similarity values show that the selected pairs are semantically related while remaining far from exact duplication. This supports the validity of the self-supervised training data construction. The maximum-comments-per-post constraint also improves diversity by preventing a small number of large posts from dominating the training pairs.

Pair mining provides the bridge between the unlabeled comment dataset and contrastive embedding training. It allows the project to improve the embedding model without manually labeled topic classes.

4.7 EMBEDDING MODEL TRAINING

Embedding model training is a central component of this project. Since the final system is an unsupervised topic modeling system and does not use manually annotated topic labels, the model was not trained as a supervised classifier. Instead, the embedding model was fine-tuned using automatically mined Comment $\leftrightarrow$ Comment positive pairs.

4.7.1 Base Encoder Selection

The base encoder used in the project is **sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2**. This model produces compact multilingual sentence embeddings with a vector dimension of 384. It is efficient enough for large-scale embedding generation while still providing useful semantic representations.

The base model is general-purpose and was not specifically trained on the project's news comment dataset. Therefore, the project fine-tunes it to better capture the semantic structure of news comments.

4.7.2 Final Model

The final adopted model is academically named the **Final Fine-Tuned MiniLM Comment Encoder**. Internally, it corresponds to experiment **V7**. The model was trained using full fine-tuning, meaning that the trainable parameters of the encoder were updated during training.

4.7.3 Training Objective

The training objective used for the final model is **MultipleNegativesRankingLoss**. This loss is suitable when positive pairs are available. For each positive pair in a batch, the corresponding paired comment is treated as the correct match, while other comments in the same batch act as implicit negatives.

The purpose of this training stage is to improve the semantic geometry of the embedding space. Related comments should become closer to each other, while unrelated comments should remain farther apart. This improved embedding space is then used by BERTopic for topic discovery.

4.7 EMBEDDING MODEL TRAINING (CONTINUED)

4.7.4 Final V7 Training Hyperparameters

The final V7 model was trained using a large balanced pair setup. The main training hyperparameters are summarized in Table 4.6.

Table 4.6: Final V7 Training Hyperparameters

Item	Value
Base model	sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2
Final model name	Final Fine-Tuned MiniLM Comment Encoder
Internal experiment name	V7
Training type	Full fine-tuning
Training pairs	1,500,000
Validation pairs	30,000
Loss function	MultipleNegativesRankingLoss
Batch size	256 pairs
Text inputs per batch	512 texts
Max sequence length	256
Learning rate	1e-5
Warmup ratio	0.10
Weight decay	0.01
Seed	42

These settings were selected to support stable contrastive fine-tuning on a large number of semantic pairs. The large batch size provides many in-batch negatives, while the sequence length of 256 allows the model to process longer comments more effectively than shorter earlier configurations.

4.8 MODEL DEVELOPMENT HISTORY

The final adopted model was reached through a sequence of experimental models. This development history is important because it shows that the final decision was not based on a single isolated experiment. Instead, each model contributed to understanding how embedding training affects downstream topic modeling.

4.8.1 Baseline Plain Model

The baseline model used the pretrained MiniLM encoder without project-specific fine-tuning. Its purpose was to provide a reference point for evaluating whether fine-tuning on news comment pairs improves downstream BERTopic behavior.

4.8.2 V1 to V4 Experimental Models

The early experimental models explored different ways to adapt the embedding model to the news comment domain. These versions helped refine the training strategy and clarify the importance of post-aware pair construction, semantic filtering, and downstream evaluation. They are treated as developmental experiments rather than final comparison candidates.

4.8.3 V5 Stage Models

The V5 family included staged fine-tuning experiments. V5 Stage 1 became an important previous strong model and is included in controlled comparisons where verified values are available. In the final report, it is referred to as **V5 Stage 1 Controlled** to distinguish its controlled evaluation values from earlier historical results.

4.8.4 V6-B Transitional Model

V6-B was a transitional full fine-tuned MiniLM model. It provided useful evidence that full fine-tuning could improve semantic separation. However, its downstream BERTopic results were weaker than the final V7 model, especially in valid topic coverage and noise reduction.

4.8 MODEL DEVELOPMENT HISTORY (CONTINUED)

4.8.5 V7 Final Fine-Tuned MiniLM Comment Encoder

V7 is the final adopted model. It uses the MiniLM backbone with full fine-tuning on 1,500,000 training pairs and 30,000 validation pairs. It was selected because it achieved the strongest downstream BERTopic balance on the fixed test pool, including the highest valid topic coverage, the lowest noise ratio, strong valid-topic confidence, and a compact topic structure.

4.8.6 V8 Additional MiniLM Experiment

V8 was an additional MiniLM experiment that used longer full fine-tuning. It achieved slightly better embedding-level retrieval metrics than V7. However, this improvement did not transfer to downstream BERTopic performance. V8 produced fewer valid topic comments, more noise, lower confidence, and lower topic diversity than V7. Therefore, it was not adopted.

4.8.7 V9 BGE Additional Backbone Experiment

V9 used a different embedding backbone based on BGE. It was tested to investigate whether changing the embedding backbone could outperform the final MiniLM-based V7 model. V9 showed reasonable embedding-level behavior, but it did not surpass V7 in downstream BERTopic evaluation.

Table 4.7 on the following page summarizes the main model variants discussed in this section, including their base architecture, training type, pair counts, and final decision status.

Table 4.7: Summary of Main Model Variants

Model	Base	Training Type	Pairs	Decision
V5 Stage 1 Controlled	MiniLM	Staged fine-tuning	Smaller pair setup	Previous strong model
V6-B	MiniLM	Full fine-tuning	31,163 train pairs	Transitional
V7 Final	MiniLM	Full fine-tuning	1,500,000 train pairs	Adopted
V8	MiniLM	Long full fine-tuning	1,500,000 train pairs	Rejected downstream
V9 BGE	BGE	Full fine-tuning	1,500,000 train pairs	Rejected downstream

The development history highlights the importance of connecting embedding training directly to downstream topic modeling evaluation. Models that appeared slightly stronger during retrieval evaluation (such as V8) did not necessarily provide the best clustering structure. V7 was chosen for achieving the most reliable balance across all topic modeling metrics.

4.9 FINAL BERTOPIC CONFIGURATION

After generating embeddings with the final fine-tuned model, the project applies BERTopic using precomputed embeddings. Therefore, BERTopic is configured with **embedding_model = None**. This ensures that the topic model uses the embeddings produced by the final trained encoder rather than generating new embeddings internally.

Table 4.8: Final BERTopic Configuration

Component	Parameter	Value
UMAP	n_neighbors / n_components	15 / 5
UMAP	min_dist / metric	0.0 / cosine
HDBSCAN	min_cluster_size	100
HDBSCAN	min_samples / metric	10 / euclidean
HDBSCAN	cluster_selection_method	eom
Vectorizer	ngram_range / min_df / max_df	(1, 2) / 2 / 0.7

This configuration was selected to produce a controlled topic structure. UMAP prepares the embedding space for clustering, HDBSCAN discovers dense topic clusters and assigns uncertain comments to noise, and c-TF-IDF extracts representative words for each topic.

4.10 IMPLEMENTATION ENVIRONMENT

The project was implemented using Python-based machine learning and natural language processing tools. The training and evaluation workflow was executed in a notebook-based environment suitable for large-scale experimentation, embedding generation, and BERTopic evaluation.

The main software components include SentenceTransformers for embedding model training, BERTopic for topic modeling, UMAP for dimensionality reduction, HDBSCAN for clustering, scikit-learn for supporting machine learning utilities, and pandas and NumPy for data processing.

Table 4.9: Main Implementation Tools

Tool / Library	Role in the Project
Python	Main programming language used for data processing and experimentation.
SentenceTransformers	Fine-tuning and using the comment embedding model.
BERTopic	Topic modeling and topic representation.
UMAP	Dimensionality reduction before clustering.
HDBSCAN	Density-based clustering and noise assignment.
pandas / NumPy	Data preparation, statistics, and result handling.

Reproducibility was supported by using a fixed random seed where applicable and by evaluating the final models on a fixed clean test pool. The final selection process also separates embedding-level analysis from downstream BERTopic evaluation, which makes the experimental reasoning clearer and more defensible.

Internal file paths and detailed saved artifacts are documented in the appendices rather than in the main methodology chapter, in order to keep the chapter readable and focused on the research process.

4.11 CHAPTER SUMMARY

This chapter presented the complete methodology of the proposed news comment topic analysis system. It described the raw comment dataset, the post-level split strategy, the text cleaning process, and the construction of clean pools for training, validation, and final evaluation.

The chapter also explained the pair mining strategy used to create Comment $\leftrightarrow$ Comment positive pairs without manual topic labels. These pairs were selected from comments belonging to the same news post and filtered using cosine similarity thresholds. The final setup used 1,500,000 training pairs and 30,000 validation pairs, making the embedding training stage a central methodological contribution.

The chapter then described the final embedding model training process. The adopted model, named the Final Fine-Tuned MiniLM Comment Encoder and internally identified as V7, was trained using MultipleNegativesRankingLoss. The objective of this training was to improve the semantic embedding space before applying BERTopic.

The model development history was also summarized. Earlier and additional models, including V5 Stage 1 Controlled, V6-B, V8, and V9 BGE, were discussed to show how the final model decision was reached. V7 was not selected merely because it was the latest model, but because it produced the strongest downstream BERTopic balance.

Finally, the chapter presented the final BERTopic configuration and the implementation environment. The next chapter presents the research results, including embedding-level analysis, downstream BERTopic results, model comparisons, rejected-model analysis, and manual topic inspection.

CHAPTER 5

RESEARCH RESULTS

This chapter presents the evaluation strategy, embedding-level results, downstream BERTopic results, final model comparisons, rejected-model analysis, manual topic inspection, and the final model selection rationale.

5.1 EVALUATION TOOLS AND STRATEGY

The evaluation strategy follows the real nature of the project. Since the system is an unsupervised topic modeling system, it is not evaluated as a supervised classifier. Therefore, final accuracy, precision, recall, and F1-score are not used as primary final metrics. Instead, the evaluation focuses on the quality of the discovered topics.

The evaluation is divided into two levels. The first level is the embedding-level evaluation. This level measures the behavior of the SentenceTransformer encoder before applying BERTopic. It includes training loss, retrieval metrics, and cosine sanity checks.

The second level is the downstream BERTopic evaluation. This level measures the final effect of the embeddings after they pass through UMAP, HDBSCAN, and c-TF-IDF. This is the most important evaluation level because the project goal is topic modeling.

Embedding-level metrics explain the behavior of the encoder. Downstream BERTopic metrics determine the final adopted model.

Table 5.1: Evaluation Levels Used in the Project

Evaluation Level	What It Measures	How It Is Used
Embedding-level	Retrieval quality and semantic separation.	Used to verify encoder learning.
BERTopic-level	Topic coverage, noise, and diversity.	Final model-selection layer.

5.2 FINAL EVALUATION DATASET

The final evaluation was conducted on a fixed clean test pool. Using a fixed evaluation dataset is essential because it ensures that model comparisons are fair. If different models are evaluated on different samples, the comparison may reflect data variation rather than real model performance.

Final evaluation pool size = **58,014 news comments**

All final BERTopic comparisons use the same number of documents. This makes it possible to compare valid topic comments, noise comments, valid ratio, noise ratio, detected topics, confidence, and topic diversity under the same evaluation conditions.

In the following section, we present the results achieved by the final adopted model (V7) when applied to this fixed evaluation dataset.

5.3 FINAL V7 BERTOPIC RESULTS

The final adopted model is the **Final Fine-Tuned MiniLM Comment Encoder**, internally referred to as **V7**. Table 5.2 summarizes its final downstream BERTopic results on the fixed test pool.

Table 5.2: Final V7 Result Summary

Metric	Value	Interpretation
Total Documents	58,014	Fixed final test pool.
Valid Topic Comments	39,458	Assigned to valid clusters.
Noise Comments	18,556	Assigned to topic -1.
Valid Ratio	68.01%	Coverage of valid topics.
Noise Ratio	31.99%	Share of noise.
Detected Topics	94	Clusters excluding noise.
Avg Confidence Valid	82.52%	Mean assignment confidence.
Topic Diversity	0.9011	Word representation diversity.

5.4 EMBEDDING-LEVEL RESULTS

Embedding-level evaluation was used to understand the behavior of the trained encoders before applying the BERTopic pipeline. These metrics analyze retrieval-side performance.

Table 5.3: Training and Retrieval-Side Metrics

Metric	V7 Final	V8	V9 BGE	Observation
Accuracy@1	0.1600	0.1622	0.1560	V8 slightly better.
Accuracy@10	0.5230	0.5330	0.5252	V8 slightly better.
MAP@100	0.2758	0.2787	0.2725	V8 slightly better.
MRR@10	0.2622	0.2654	0.2592	V8 slightly better.
NDCG@10	0.3239	0.3286	0.3220	V8 slightly better.

The table shows that V8 achieved slightly stronger retrieval-side metrics than V7. This proves that the final decision was not made by blindly selecting the model with the best training-side values, as V8 did not produce better downstream topic modeling.

5.5 POSITIVE-VS-RANDOM COSINE SANITY CHECK

The cosine sanity check compares the average similarity of positive pairs with the average similarity of random pairs.

Table 5.4: Positive-vs-Random Pair Sanity Check

Model	Positive Mean	Random Mean	Gap
V6-B	0.4812	0.0698	0.4114
V7 Final	0.5556	0.0728	0.4828
V8	0.5523	0.0479	0.5044
V9 BGE	0.5579	0.0448	0.5131

Both V8 and V9 showed strong separation, but a stronger sanity gap does not automatically guarantee better downstream BERTopic output.

5.6 COMPARISON BETWEEN PROPOSED MODELS

The main model comparison is based on downstream BERTopic evaluation, measuring the final goal: producing useful topic structures.

Table 5.5: BERTopic Model Comparison (Fixed Test Pool)

Metric	V5 S1	V6-B	V7 Final	V8	V9 BGE
Valid Comments	38,855	37,134	39,458	37,504	37,581
Noise Comments	19,159	20,880	18,556	20,510	20,433
Valid Ratio	66.98%	64.01%	68.01%	64.65%	64.78%
Noise Ratio	33.02%	35.99%	31.99%	35.35%	35.22%
Detected Topics	111	104	94	110	116

V7 produced the highest valid topic coverage and achieved the lowest noise ratio, making it the strongest downstream balance.

5.7 CONFIDENCE AND DIVERSITY COMPARISON

Table 5.6: Confidence and Topic Diversity Comparison

Metric	V5 S1	V6-B	V7 Final	V8	V9 BGE
Avg Conf. All	55.22%	51.73%	56.13%	33.46%	33.66%
Avg Conf. Valid	82.45%	80.82%	82.52%	49.72%	50.04%
Med. Conf. Valid	98.96%	93.50%	98.08%	33.82%	32.91%
Topic Diversity	0.9036	0.9048	0.9011	0.8755	0.8784

V7 achieved the highest average valid confidence. Although V6-B produced a slightly higher diversity score, this isolated advantage does not outweigh V7's superior coverage and noise reduction.

5.8 DIRECT COMPARISON AGAINST V7

Table 5.7: Models Compared Directly Against V7

Model	Valid Diff.	Noise Diff.	Noise Ratio Diff.	Avg Conf. Diff.
V5 Stage 1	-603	+603	+1.03 pts	-0.07 pts
V6-B	-2,324	+2,324	+4.00 pts	-1.70 pts
V8	-1,954	+1,954	+3.36 pts	-32.80 pts
V9 BGE	-1,877	+1,877	+3.23 pts	-32.48 pts

The comparison confirms that V7 assigned additional comments to valid topics and reduced noise significantly compared to its predecessors and rejected variants.

5.9 WHY SOME MODELS WERE NOT ADOPTED

5.9.1 V5 Stage 1 Controlled

Produced fewer valid topic comments and a higher noise ratio than V7. Its topic structure was also less compact.

5.9.2 V6-B

Weaker in downstream coverage, producing 2,324 fewer valid comments than V7.

5.9.3 V8

Improved retrieval metrics did not transfer to clustering. It saw a massive drop in valid-topic confidence.

5.9.4 V9 BGE

Strong sanity separation but did not outperform V7 in topic modeling coverage or confidence.

V7 was adopted because it achieved the strongest downstream BERTopic balance, not because it had the lowest training loss.

5.10 COMPARISON WITH PREVIOUS WORKS

The comparison with previous work is qualitative. The proposed system contributes a controlled pipeline specific to the domain of news comments.

Table 5.8: Qualitative Comparison with Related Work Directions

Direction	Strength	Limitation
Classical modeling	Interpretable.	Weak for short comments.
General BERTopic	Effective modeling.	Depends on embedding quality.
LLM-assisted	Clearer labeling.	Cost and dependency.

This project improves upon general BERTopic usage by fine-tuning the embeddings specifically for short-text public discourse before clustering.

5.11 MANUAL TOPIC INSPECTION

5.11.1 Strong Topic Examples

Strong topics are those whose representative words form a clear and coherent theme.

Table 5.9: Strong Topic Examples

Theme	Representative Words	Interpretation
Russia	russia, putin, russian	Geopolitics.
Healthcare	healthcare, insurance, medical	Health policy.
Religion	christian, church, religion	Social/Religious.
Election	vote, voter, election	Politics/Voting.
Military	war, military, nuclear, army	Security/War.

5.11 MANUAL TOPIC INSPECTION (CONTINUED)

5.11.2 Weak and Mixed Topic Examples

Table 5.10: Weak and Mixed Topic Examples

Type	Representative Words	Interpretation
Food / Generic	food, chocolate, sugar, burger	Medium-quality.
Reaction-Based	late, surprise, awesome	Weak topic.
Name-Based	chris, david, sean, michael	Weak topic.

These weak examples reflect the nature of the dataset. News comments often contain generic reactions and jokes that do not form a single semantic theme.

The presence of some weak topics is expected. Final evaluation considers the overall topic structure, not just isolated examples.

5.12 STRENGTHS AND WEAKNESSES

5.12.1 Strengths

- Unsupervised: Does not require manual labels.
- Data integrity: Post-level split reduces leakage.
- Semantic focus: Contrastive fine-tuning adapts the encoder.
- Downstream optimized: Selection based on BERTopic behavior.

5.12.2 Weaknesses

- No absolute ground-truth for unsupervised metrics.
- Persistence of generic/noisy clusters in short-text data.
- Sensitivity to UMAP/HDBSCAN parameter selection.
- Manual inspection remains partially subjective.

5.13 FINAL MODEL SELECTION RATIONALE

V7 was selected because it achieved the strongest downstream balance on the fixed pool. It produced the highest number of valid topic comments and the lowest noise ratio (31.99%). Rejected variants (V8, V9) showed that training-side improvement does not always lead to better clustering.

5.14 CHAPTER SUMMARY

This chapter presented the results on a fixed pool of 58,014 comments. V7 produced 39,458 valid topic comments and 94 detected topics with an average confidence of 82.52%. Comparisons confirmed that V7 outperformed all other variants in valid coverage and noise reduction. Manual inspection verified the presence of strong interpretable themes.

The next chapter concludes the report by summarizing findings and presenting future work directions.

CHAPTER 6

CONCLUSION

This chapter summarizes the main conclusions of the project, reviews the achieved contributions, discusses the main technical challenges, and presents future work directions for improving the proposed news comment topic analysis system.

6.1 CONCLUSIONS

This project presented an unsupervised topic modeling system for analyzing large-scale news comments. The main objective was to discover coherent discussion topics from noisy, short, and informal user-generated comments without relying on manually labeled topic classes. Therefore, the project was not treated as a supervised classification task, and the final evaluation did not depend on accuracy, precision, recall, or F1-score as primary metrics.

The proposed solution used a fine-tuned SentenceTransformer model followed by a BERTopic-based pipeline. The embedding model was trained using automatically mined Comment $\leftrightarrow$ Comment positive pairs rather than human-defined topic labels. This training strategy allowed the system to improve the semantic embedding space before applying dimensionality reduction, clustering, and topic representation.

The final adopted model was the **Final Fine-Tuned MiniLM Comment Encoder**, internally referred to as **V7**. It was trained using **1,500,000** positive training pairs and **30,000** validation pairs with **MultipleNegativesRankingLoss**. Its embeddings were then used inside the BERTopic pipeline through UMAP, HDBSCAN, and c-TF-IDF.

The final evaluation was performed on a fixed clean test pool of **58,014** news comments. The final model assigned **39,458** comments to valid topic clusters and **18,556** comments to noise. It achieved a **Valid Ratio of 68.01%**, a **Noise Ratio of 31.99%**, **94 detected topics**, an **Average Confidence Valid of 82.52%**, and a **Topic Diversity score of 0.9011**.

The main conclusion is that improving the semantic embedding model before applying BERTopic helped reduce noise, increase valid topic assignment, and produce a more controlled topic structure for news comment analysis.

6.2 PROJECT CONTRIBUTIONS

The project provides several methodological and practical contributions. First, it presents a complete unsupervised topic modeling pipeline for news comments. The pipeline starts from raw comments, applies cleaning and filtering, preserves a post-level split, constructs semantic training pairs, fine-tunes an embedding model, generates embeddings, applies BERTopic, and evaluates the discovered topics.

Second, the project uses a post-level split strategy to reduce data leakage. This is an important methodological decision because comments under the same news post often share contextual meaning. Keeping all comments from the same post within the same split makes the evaluation more realistic and prevents the model from indirectly seeing related discussion patterns during training.

Third, the project constructs training data without manually labeled topic classes. Instead of assigning predefined labels to comments, it uses automatically mined Comment $\leftrightarrow$ Comment positive pairs. These pairs were selected from comments belonging to the same post and filtered using cosine similarity thresholds. This allowed the project to perform contrastive embedding training while preserving the unsupervised nature of the final topic modeling task.

Fourth, the project separates embedding-level evaluation from downstream BERTopic evaluation. Training loss, retrieval metrics, and cosine sanity checks were used to understand encoder behavior, while downstream BERTopic metrics were used for final model selection. This separation made the evaluation more precise and prevented the incorrect assumption that lower training loss automatically produces better topic modeling.

Fifth, the project compared the adopted model with previous and additional models, including V5 Stage 1 Controlled, V6-B, V8, and V9 BGE. These comparisons showed that V7 achieved the strongest downstream BERTopic balance, even when some other models achieved better isolated embedding-side indicators.

6.3 CHALLENGES AND LIMITATIONS

Despite the positive results, the project has several limitations. The first limitation is the nature of news comments themselves. Many comments are short, informal, sarcastic, emotional, or context-dependent. Some comments do not contain enough information to be assigned confidently to a coherent topic. This naturally increases the difficulty of topic modeling and explains why a portion of comments remains assigned to noise.

The second limitation is the unsupervised nature of the task. Since the dataset does not contain manually verified topic labels, there is no single ground-truth metric that can fully determine topic quality. Metrics such as noise ratio, valid topic comments, confidence, detected topics, and topic diversity are useful, but they must be interpreted together. Manual topic inspection remains necessary.

The third limitation is the sensitivity of BERTopic to configuration choices. UMAP and HDBSCAN parameters can affect the number of topics, the noise ratio, and the structure of clusters. A configuration that reduces noise may sometimes produce weaker or more mixed topics, while a more conservative configuration may increase noise but improve topic purity.

The fourth limitation is that topic diversity should not be interpreted alone. A model may achieve a slightly higher diversity score while producing more noise or assigning fewer comments to valid topics. For this reason, the project selected the final model using the complete downstream BERTopic balance rather than one isolated metric.

Finally, manual inspection is partially subjective. Although it is necessary for unsupervised topic modeling, future work should expand manual evaluation using clearer annotation guidelines, multiple reviewers, or agreement-based evaluation when possible.

6.4 FUTURE WORK

Future work can improve the project in several directions. The first direction is to expand the evaluation process. Manual topic inspection can be made more rigorous by using multiple reviewers, predefined topic-quality criteria, and agreement measurements. This would make qualitative evaluation more reliable and less dependent on a single reviewer.

The second direction is to improve multilingual and Arabic-aware topic modeling. Since news comments may include Arabic, English, mixed language, informal expressions, and dialectal forms, future versions of the system can test additional multilingual or Arabic-focused embedding models. This may improve topic quality for non-English or mixed comment datasets.

The third direction is to improve topic labeling and summarization. The current system uses representative words extracted through c-TF-IDF. Future work can add an optional interpretation layer that generates clearer topic labels and summaries. Large language models may be useful for this purpose, but they should be used carefully to preserve reproducibility and avoid replacing the core unsupervised topic modeling pipeline.

The fourth direction is to extend temporal analysis. News comments are often connected to evolving events. Adding topic-over-time analysis would allow the system to track how public discussions change across dates, events, or news cycles.

The fifth direction is to improve the user interface. A stable dashboard can help users explore topics, inspect representative comments, filter by time or topic, export results, and compare model outputs. This would make the system more useful for researchers, media analysts, and decision makers.

6.5 RECOMMENDATIONS

The main recommendation is to treat the system as an analytical support tool rather than a fully automatic replacement for human interpretation. The system can organize large volumes of comments, reduce manual workload, and reveal recurring discussion themes, but final interpretation should still involve human review.

6.6 CLOSING REMARKS

This project demonstrates that improving the embedding space before applying topic modeling can strengthen the analysis of noisy news comments. The final model, V7, was selected because it achieved the best downstream BERTopic balance, not because it had the lowest training loss or the strongest isolated retrieval metric.

The experiments showed that embedding-level improvements do not always transfer directly to better topic modeling. V8 achieved slightly stronger retrieval-side indicators, and V9 showed strong positive-random separation, but both produced weaker BERTopic results than V7. This confirms the importance of evaluating models according to the final task rather than only according to training-side indicators.

The final system provides a structured method for discovering topics from large-scale news comments without manually labeled topic classes. It combines post-level split, automatic pair mining, contrastive fine-tuning, semantic embeddings, UMAP, HDBSCAN, BERTopic, c-TF-IDF, numerical evaluation, and manual inspection.

The final adopted model is the Final Fine-Tuned MiniLM Comment Encoder, internally corresponding to V7. It is selected based on downstream BERTopic evaluation on the fixed test pool of 58,014 news comments.

Overall, the project offers a practical and academically defensible pipeline for unsupervised topic analysis of news comments. It also establishes a clear foundation for future improvements in multilingual modeling, topic labeling, temporal analysis, and interactive exploration.

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The references above support the theoretical background, related work, methodology, and implementation choices used in the proposed system. Internal project artifacts, experiment outputs, and saved model paths are documented separately in the appendices.

APPENDIX A

FULL PARAMETER SUMMARY

This appendix summarizes the main dataset, cleaning, pair mining, model training, BERTopic configuration, and evaluation parameters used in the project. It supports reproducibility without overloading the main chapters.

APPENDIX A.1 DATASET AND CLEANING PARAMETERS

Table A.1 summarizes the main dataset structure and filtering outputs used before embedding generation.

Table A.1: Dataset Summary

Item	Value	Meaning
Raw comments shape	$1,038,319 \times 5$	Original dataset before filtering.
Main text column	message	Textual field used for modeling.
Post split shape	$17,351 \times 4$	Used for post-level separation.
Before filtering	1,035,495	Comment count at start.
After filtering	843,636	Total clean comments available.
Removed comments	191,859	Entries filtered as noise/empty.

Table A.2: Split Distribution After Filtering

Split	Count	Usage
Train	593,543	Pair mining and training.
Validation	125,614	Validation metrics.
Test	124,479	Hold-out pool.
Fixed Evaluation Pool	58,014	Final model comparison pool.

APPENDIX A.2 PAIR MINING PARAMETERS

Final training examples are Comment $\leftrightarrow$ Comment positive pairs filtered using cosine similarity.

Table A.3: Pair Mining Configuration

Parameter	Training	Validation
Comment pool	593,543	125,614
PAIR_MIN_SIM	0.45	0.45
PAIR_MAX_SIM	0.90	0.90
MAX_COMMENTS / POST	200	200
MAX_PAIRS / POST	5,000	1,500
Candidate pairs	3,701,173	752,789
Final selected pairs	1,500,000	30,000
Mean pair similarity	0.6024	0.6075

Pair mining provided the semantic training signals needed for contrastive learning without manual topic labels.

APPENDIX A.3 MODEL DEVELOPMENT SUMMARY

Table A.4: Model Development Summary

Model	Base	Pair Setup	Final Role
Baseline	MiniLM	None	Reference baseline.
V1-V4	MiniLM	Limited	Historical development stages.
V5 S1	MiniLM	Staged	Previous strong model (Controlled).
V6-B	MiniLM	31k pairs	Transitional full fine-tuning.
V7 Final	MiniLM	1.5M pairs	Adopted model.
V8	MiniLM	1.5M pairs	Rejected (Downstream noise).
V9	BGE	1.5M pairs	Rejected (Backbone experiment).

The early V1 to V4 models are recorded as historical development stages and were not treated as final controlled competitors.

APPENDIX A.4 TRAINING PARAMETERS BY MAIN MODEL

Table A.5: Training Parameters by Main Model

Model	Type	Train Pairs	Batch	Seq	Loss
V5 S1	Staged	31,163	32	96	Pair/Triplet
V6-B	Full FT	31,163	128	256	MNRL
V7 Final	Full FT	1,500,000	256	256	MNRL
V8	Full FT	1,500,000	256	256	MNRL
V9 BGE	Full FT	1,500,000	128	256	MNRL

MNRL: MultipleNegativesRankingLoss. Full FT: Full Fine-Tuning of all encoder parameters.

Appendix A.4.1 Final V7 Training Configuration

Parameter	Final V7 Value	Purpose
Learning rate	1e-5	Stable fine-tuning.
Warmup ratio	0.10	LR stability.
Weight decay	0.01	Regularization.
Seed	42	Reproducibility.

These settings supported stable contrastive fine-tuning on the large-scale pair setup, allowing the encoder to adapt effectively to news comment semantics.

APPENDIX A.5 FINAL BERTOPIC CONFIGURATION

Table A.6: Final BERTopic Pipeline Configuration

Component	Parameter	Value
Embeddings	embedding_model	None (Precomputed)
UMAP	n_neighbors	15
UMAP	n_components	5
UMAP	metric	cosine
HDBSCAN	min_cluster_size	100
HDBSCAN	min_samples	10
HDBSCAN	metric	euclidean
HDBSCAN	cluster_selection	eom
Vectorizer	ngram_range	(1, 2)
Vectorizer	min_df / max_df	2 / 0.7

APPENDIX A.6 EVALUATION METRICS SUMMARY

Table A.7: Embedding-Level Metrics

Metric	Used For	Role
Training Loss	Optimization monitor.	Diagnostic.
Validation Loss	Training check.	Diagnostic.
Accuracy@K	Retrieval behavior.	Secondary.
MAP / MRR / NDCG	Ranking quality.	Secondary.
Cosine Gap	Semantic separation.	Sanity check.

Table A.8: Downstream BERTopic-Level Metrics

Metric	Used For	Goal
Valid Comments	Topic coverage.	Higher.
Noise Ratio	Unassigned share.	Lower.
Detected Topics	Cluster count.	Balanced.
Avg Conf Valid	Mean confidence.	Higher.
Topic Diversity	Word diversity.	Higher.

Final model selection was based on downstream BERTopic performance on the fixed test pool, not on training loss alone.

APPENDIX B

FULL EMBEDDING-LEVEL COMPARISONS

This appendix documents the embedding-level evaluation of the trained encoders. It includes training loss, retrieval metrics, precision and recall indicators, cosine sanity checks, and the interpretation of V7, V8, and V9.

APPENDIX B.1 PURPOSE OF EMBEDDING-LEVEL EVALUATION

Embedding-level evaluation measures the behavior of the trained encoder before it is used inside the BERTopic pipeline. This level is essential because the embedding model is responsible for transforming each comment into a semantic vector.

Table B.1: Evaluation Role of Embedding-Level Metrics

Metric Family	What It Explains	Final Role
Loss metrics	Optimization behavior during fine-tuning.	Diagnostic.
Retrieval metrics	Semantic proximity of related comments.	Secondary evidence.
Precision / Recall	Top-K retrieval-side behavior.	Training analysis.
Cosine sanity	Positive-vs-random pair separation.	Sanity validation.

Embedding metrics verify whether the encoder learned useful similarity, but they do not replace downstream evaluation as the final decision layer.

APPENDIX B.2 TRAINING LOSS CHECKPOINTS

Table B.2 shows the optimization behavior. Lower loss indicates better training-side optimization but does not automatically imply better clustering results.

Table B.2: Training Loss Checkpoints

Model	Checkpoint	Step	Train Loss	Eval Loss
V7 Final	checkpoint-4000	4,000	1.5707	1.6494
V7 Final	checkpoint-5860	5,860	1.5350	1.6494
V8	checkpoint-16000	16,000	1.3511	1.6152
V8	checkpoint-22000	22,000	1.2621	1.6152

V8 achieved lower losses than V7 due to longer fine-tuning, but this did not transfer into improved downstream BERTopic coverage or confidence.

APPENDIX B.3 INFORMATION RETRIEVAL METRICS

IR metrics measure how well the model retrieves related comments. These were used to verify semantic similarity behavior before topic modeling.

Table B.3: Main Retrieval Metrics Comparison

Metric	V7 Final	V8	V9 BGE
Accuracy@1	0.1600	0.1622	0.1560
Accuracy@10	0.5230	0.5330	0.5252
MAP@100	0.2758	0.2787	0.2725
MRR@10	0.2622	0.2654	0.2592
NDCG@10	0.3239	0.3286	0.3220

V8 is slightly stronger on retrieval metrics, confirming that it learned a stronger embedding-level retrieval signal, which nonetheless produced more noise in BERTopic.

APPENDIX B.4 PRECISION AND RECALL METRICS

Table B.4: Precision and Recall @K

Metric	V7 Final	V8	V9 BGE
Precision@1	0.1600	0.1622	0.1560
Precision@5	0.0809	0.0812	0.0791
Recall@1	0.1600	0.1622	0.1560
Recall@10	0.5230	0.5330	0.5252

Precision and recall results follow the same pattern as Accuracy and MAP, with V8 showing marginal training-side improvements over the adopted V7 model.

APPENDIX B.5 POSITIVE-VS-RANDOM COSINE SANITY CHECKS

A useful model must assign higher similarity to related pairs than to random pairs.

Table B.6: Positive-vs-Random Cosine Gap

Model	Pos Mean	Rand Mean	Gap
V6-B	0.4812	0.0698	0.4114
V7 Final	0.5556	0.0728	0.4828
V8	0.5523	0.0479	0.5044
V9 BGE	0.5579	0.0448	0.5131

V9 achieved the strongest gap, showing excellent separation. However, this did not translate to better clustering coverage compared to V7.

APPENDIX B.6 V9 BGE RETRIEVAL PROGRESS

Table B.7: V9 Retrieval Progress Across Training Steps

Step	Acc@1	Acc@3	Acc@5	Acc@10	MRR@10	NDCG@10	MAP@100
2,000	0.1428	0.2862	0.3686	0.4836	0.2386	0.2965	0.2517
8,000	0.1558	0.3088	0.3868	0.5166	0.2566	0.3179	0.2695
16,000	0.1560	0.3102	0.3956	0.5252	0.2592	0.3220	0.2725
22,000	0.1566	0.3058	0.3978	0.5298	0.2596	0.3233	0.2727

Progress shows consistent learning across steps for V9. However, the final gains were insufficient to surpass the downstream performance of the MiniLM-based V7 model.

APPENDIX B.7 FINAL INTERPRETATION

Table B.8: Final Interpretation of Embedding-Level Models

Model	Embedding Behavior	Interpretation
V7 Final	Strong and stable.	Adopted; Best balance.
V8	Slightly stronger IR.	Rejected; Downstream noise.
V9 BGE	Reasonable BGE behavior.	Rejected; Backbone exp.

APPENDIX B.8 SUMMARY

The embedding-level results show that while V8 and V9 achieved isolated training-side advantages, they did not improve downstream clustering. V7 remains the adopted model as it achieved the best downstream BERTopic balance on the fixed test pool.

APPENDIX C

ADDITIONAL BERTOPIC COMPARISONS

This appendix provides additional downstream BERTopic comparisons, rankings, and direct model differences. These results support the final decision to adopt V7 as the Final Fine-Tuned MiniLM Comment Encoder.

APPENDIX C.1 FINAL BERTOPIC COMPARISON

Table C.1 presents the final downstream BERTopic comparison across the main evaluated models on the same fixed test pool of 58,014 news comments.

Table C.1: Final BERTopic Comparison (Fixed Test Pool)

Metric	V5 S1	V6-B	V7 Final	V8	V9 BGE
Valid Comments	38,855	37,134	39,458	37,504	37,581
Noise Comments	19,159	20,880	18,556	20,510	20,433
Valid Ratio	66.98%	64.01%	68.01%	64.65%	64.78%
Noise Ratio	33.02%	35.99%	31.99%	35.35%	35.22%
Detected Topics	111	104	94	110	116

V7 assigned the largest number of comments to valid topics and produced the lowest noise ratio while maintaining a compact topic structure.

APPENDIX C.2 RANKINGS BY COVERAGE AND NOISE

Table C.2: Ranking by Valid Topic Comments

Rank	Model	Count	Reading
1	V7 Final	39,458	Highest valid topic assignment.
2	V5 S1	38,855	Strong previous model.
3	V9 BGE	37,581	Reasonable performance.
4	V8	37,504	Lower coverage.
5	V6-B	37,134	Lowest coverage.

Table C.3: Ranking by Noise Ratio

Rank	Model	Ratio	Reading
1	V7 Final	31.99%	Lowest noise ratio.
2	V5 S1	33.02%	Close to V7 but noisier.
3	V9 BGE	35.22%	Higher noise.
4	V8	35.35%	Higher noise despite IR metrics.
5	V6-B	35.99%	Highest noise ratio.

APPENDIX C.3 CONFIDENCE AND DIVERSITY RANKINGS

Table C.4: Ranking by Avg Valid-Topic Confidence

Rank	Model	Confidence	Reading
1	V7 Final	82.52%	Highest mean confidence.
2	V5 S1	82.45%	Very close to V7.
3	V6-B	80.82%	Stable confidence.
4	V9 BGE	50.04%	Significantly lower.
5	V8	49.72%	Lowest confidence.

Table C.5: Ranking by Topic Diversity

Rank	Model	Diversity	Reading
1	V6-B	0.9048	Highest diversity score.
2	V5 S1	0.9036	Strong diversity.
3	V7 Final	0.9011	Stable diversity (Adopted).
4	V9 BGE	0.8784	Reduced diversity.
5	V8	0.8755	Lowest diversity.

APPENDIX C.4 DIRECT DIFFERENCES AGAINST V7

Negative valid-topic difference means fewer valid comments than V7. Positive noise difference means more noise comments than V7.

Table C.6: Direct Comparison Against V7

Model	Valid Diff.	Noise Diff.	Ratio Diff.	Conf. Diff.
V5 S1	-603	+603	+1.03 pts	-0.07 pts
V6-B	-2,324	+2,324	+4.00 pts	-1.70 pts
V8	-1,954	+1,954	+3.36 pts	-32.80 pts
V9 BGE	-1,877	+1,877	+3.23 pts	-32.48 pts

V8 and V9 suffered a large drop in confidence despite reasonable embedding metrics, confirming that IR improvement does not imply clustering success.

APPENDIX C.5 FINAL BERTOPIC INTERPRETATION

The additional comparisons confirm that V7 provides the best overall BERTopic behavior. It achieves the optimal balance between valid topic coverage, noise reduction, and average confidence.

V5 Stage 1 Controlled remains a strong historical comparison, but it assigns fewer comments to valid topics and produces more noise than the final adopted V7 model. V6-B, while showing higher diversity, remains weaker in coverage and noise reduction.

The rejection of V8 and V9 is structurally important for the methodology. It validates that final model selection must be based on the downstream task—topic modeling—rather than training-side indicators or backbone families alone.

Final decision: V7 is adopted because it achieves the strongest downstream BERTopic balance on the fixed evaluation pool of 58,014 comments.